AuditGuard Complete User Manual

Installation, dependencies, startup, every feature, testing, repeat usage, industry value, and troubleshooting

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This is the complete AuditGuard operating manual. It includes prerequisites, installation, dependency setup, first-time and repeat startup, every feature and tab, testing, troubleshooting, industry use, and demo presentation steps.

1. What must be installed before running the code

Required software

Windows 10/11, Python 3.10 or newer, PowerShell, and the AuditGuard project folder. Git is not required for normal use after the project folder is available. An internet connection is only needed to install packages or use optional Groq explanations.

Python check

Open PowerShell and run:

```
python --version py --version
```

If Python is not found, install Python from python.org and enable ?Add Python to PATH?. Close and reopen PowerShell after installation.

Python packages installed by requirements.txt

The project uses: streamlit, pandas, numpy, plotly, openpyxl, scipy, jinja2, requests, python-dotenv, and pytest. These provide the web dashboard, data processing, charts, Excel/CSV handling, statistical Benford testing, optional .env support, and automated tests.

2. Complete installation from zero

Run these commands in PowerShell:

```
cd "D:\AI Audit Red-Flag Dashboard\auditguard" py -m venv ..\.venv ..\.venv\Scripts\activate python -m pip install --upgrade pip pip install -r requirements.txt
```

If the project already contains .venv, do not create it again; activate the existing environment and run pip install -r requirements.txt to restore dependencies.

Verify installation:

```
python -c "import streamlit, pandas, numpy, plotly, openpyxl, scipy, requests; print('AuditGuard dependencies OK')"
```

3. Start, stop, and restart

Start the dashboard:

```
cd "D:\AI Audit Red-Flag Dashboard\auditguard" ..\.venv\Scripts\activate python -m streamlit run app.py
```

Open http://localhost:8501 if the browser does not open automatically. Keep the terminal open while using the app. Stop safely by focusing that terminal and pressing Ctrl+C. To use another port: python -m streamlit run app.py --server.port 8502.

4. First-time use - exact steps

1. Start Streamlit. 2. Open the browser page. 3. Confirm the empty landing state and required-column list. 4. Press Load Demo Data, or upload an .xlsx, .xlsm, or .csv file. 5. Confirm the eight required columns. 6. Map missing columns if the mapping panel appears. 7. Configure date format, approval limit, round threshold, weekends, holidays, financial year, number format, and optional AI. 8. Press Run Audit. 9. Review warnings/errors. 10. Visit each tab and download the report.

5. Second-time use

If Streamlit is still running, simply upload a new file or refresh the browser. Review the settings because they are session settings, then press Run Audit again. Use Load Demo Data to return to the sample. If the server was stopped, activate the environment and run the start command again. Uploaded files, audit results, and session keys are temporary and disappear

when the session is refreshed or restarted.

6. Run the automated tests

From the auditguard folder with the virtual environment active, run:

```
python -m pytest -q
```

The tests cover detection rules, input validation, data generation, AI explanations, Groq safety, calendar rendering, exports, UI helpers, and Streamlit smoke behavior. A successful run should finish without test failures.

7. Optional Groq setup

Groq is not required for the audit engine. The app works fully with deterministic explanations. To use Groq, open the sidebar Groq API keys section, paste one or more keys, press Add key, test the key, choose a model, and enable Use LLM explanations. Keys may also be supplied through GROQ_API_KEYS in a local .env file. Never commit a real key to source control.

8. Input file requirements

Required columns: EntryID, Date, Account, Vendor, Description, Amount, PostedBy, ApprovedBy. Optional GroundTruth enables accuracy reporting. Excel reads the Ledger sheet when present; otherwise the first sheet. CSV supports comma, semicolon, and tab delimiters. Dates can be Auto-detect, MM/DD/YYYY, or DD/MM/YYYY. Amounts must be numeric and non-negative.

9. Complete feature guide

AuditGuard — Feature Guide

This guide explains every feature in simple English, with a short example for

each one. The same explanations are available inside the app on the **Guide**

tab.

1. What AuditGuard does

AuditGuard takes a journal ledger and turns it into a ranked review list.

1. Read the ledger and check the eight required columns.
2. Run six audit tests on every row.
3. Give each row a risk score from 0 to 100.
4. Show why each flagged row was flagged.
5. Plot risky days on a calendar.
6. Export an Excel audit pack and, when labels exist, measure accuracy.

Everything happens inside the browser session. There is no database, and

uploaded files are gone when the session ends.

Example: a 606-row 2025 ledger is scored in under a second and produces 57

flagged entries, Rs. 77,29,596.13 of amount at risk, and an overall High risk

score of 47.3.

2. Loading data

Upload a file

Use the sidebar uploader for `.xlsx`, `.xlsm`, or `.csv`.

? Required columns: `EntryID`, `Date`, `Account`, `Vendor`, `Description`, `Amount`, `PostedBy`, `ApprovedBy`.
? Optional: `GroundTruth` — one of `Duplicate`, `RoundAmount`, `Weekend`, `JustBelowLimit`, `SelfApproval`, `BenfordManip`, `MultiRule`, or `Clean`.
? Excel workbooks use the `Ledger` sheet when it exists; otherwise the first sheet.
? CSV files may use commas, semicolons, or tabs.

Example: the blind demo workbook has only a `Ledger` sheet. Upload it and press **Run Audit**; the app does not need `GroundTruth` to score the ledger.

Column mapping

If a required column is missing, AuditGuard opens a mapping panel. Pick a source column for each missing field, then press **Apply mapping and run audit**. Duplicate selections are blocked.

Example: a file with `Supplier` instead of `Vendor` can be mapped in one click; the app suggests `Supplier` automatically.

Friendly problems

? Missing columns → mapping panel.
? Unreadable dates or amounts → row-level warnings, not a crash.
? Negative amounts, no usable dates, or fewer than two rows → a clear error explains what to fix.

3. Money and number formats

All amounts are Pakistani rupees.

Whole amounts have no decimals; fractional amounts keep two decimals. The same formatter is used in the dashboard, explanations, CSV, and Excel, so the number never changes shape between views.

Example: an amount of `1_250_000.50` appears as `Rs. 12,50,000.50` in the table and as `Rs. 12.5 Lac` in a compact KPI.

4. Dates and Karachi time

Every displayed date is `MM/DD/YYYY`. Reports and audit timestamps use

Asia/Karachi and show `PKT`.

? **Input date format:** Auto-detect, MM/DD/YYYY (default), or DD/MM/YYYY.
? When auto-detect sees a date that could be either day-first or month-first, it warns and lists affected rows instead of silently guessing.
? The holiday box accepts `MM/DD/YYYY` and `YYYY-MM-DD`.
? Calendar day inspectors, Plotly hover text, CSV, and Excel all use the same format.

Example: `03/04/2025` is 4 March 2025 under MM/DD/YYYY and 3 April 2025 under DD/MM/YYYY. Choose the format that matches the source file.

5. Karachi and Pakistan settings

The sidebar's **Where do these holiday dates come from?** section spells out each source: the seven fixed dates come from AuditGuard's built-in Pakistan calendar; the extra dates are whatever you type in the box (or the four moon-sighting dates stored in the demo workbook).

Fixed holidays:

? 02/05 Kashmir Day
? 03/23 Pakistan Day
? 05/01 Labour Day
? 05/28 Youm-e-Takbeer
? 08/14 Independence Day
? 11/09 Iqbal Day
? 12/25 Quaid-e-Azam Day

Example: a posting on Saturday 03/29/2025 is flagged under the default weekend. Switching to **Sunday only** stops that flag, but a posting on 08/14/2025 stays flagged because it is Independence Day. Pick **Custom** weekend days to mirror a six-day week, a Friday–Saturday weekend, or any other office calendar.

6. The six audit tests

Duplicate

Same vendor and amount within seven days.

? **Both** postings are flagged.
? Each row records the other row's `EntryID` and the day gap.
? If both descriptions share an invoice reference such as `INV-2025-0142`, the reason mentions it as stronger evidence.

? The seven-day rule and the ledger schema do not change.

Example: `Indus Air Services` at `Rs. 3,918.37` on 03/12/2025 and again on 03/17/2025 → both rows flagged as Duplicate.

Round amount

Exact multiples of `Rs. 1,000` at or above `Rs. 10,000`.

Example: `Rs. 75,000` is flagged; `Rs. 75,250` is not.

Weekend or holiday

A posting on a configured weekend day or a Pakistan public holiday.

Example: Saturday 11/01/2025 → Weekend Posting. Friday 10/31/2025 → no weekend flag.

Just below limit

An amount from 95% to 99.99% of the approval limit.

Example: with the default `Rs. 5,00,000` limit, `Rs. 4,87,500` (97.5%) is flagged; `Rs. 3,00,000` is not.

Self-approval

The same person posted and approved the entry, compared after trimming spaces and ignoring upper/lower case.

Example: `PostedBy = Hamza Sheikh` and `ApprovedBy = hamza sheikh` → Self-Approval.

Benford deviation

Vendors with at least 15 entries are tested with a first-digit chi-square against Benford's Law. If `p < 0.01`, every entry from that vendor is flagged for review.

Example: `Madina Traders` has 18 fabricated entries whose leading digits are 5–9; the vendor-level test flags all 18.

Benford is a sampling pointer, not proof of fraud. Rent entries from `Pearl Property Rentals` are excluded from digit testing so fixed monthly rent does not distort the result.

7. Risk scoring

Points add up and are capped at 100.

The portfolio score is `0.7 × average flagged score + 3 × flagged percentage`, capped at 100. **Amount at risk** is the total absolute value of flagged entries.

Example: Self-Approval + Weekend + Just Below Limit = 35 + 10 + 25 = 70 → Critical.

The weights and the round threshold can be changed under **Advanced settings**. Re-run the audit after changing them.

8. Overview tab

? Four KPI cards: total entries, flagged entries and percentage, amount at risk, and overall risk score/level.
? Rule-hit bar chart and risk-mix donut.
? Flagged timeline.
? Benford expected-vs-actual chart with the p-value.
? Top vendors by flagged exposure.
? A short methodology panel with the current thresholds.

Example: on the demo the Benford rule produces the most hits (18), followed by duplicates (12) and self-approvals (11).

9. Flagged Entries tab

Filters:

? risk level,
? flag type,
? vendor,
? minimum risk score,
? date range,
? description search.

The table is sorted by risk score. Select any Entry ID to see every field, the reasons, the full explanation, recommended audit procedures, and the related duplicate row where applicable.

Example: choose **Self-Approval** → 11 of 57 entries. Add **Critical** under risk level → 2 entries remain.

Dates in the inspector and table use `MM/DD/YYYY`; amounts use `Rs.`.

10. AI Insights tab

? Executive summary built from the audit numbers.

? Top-N entry cards with explanations and procedures.

? A selector to explain any flagged entry.

? **English / Roman Urdu** toggle.

? Optional Groq wording per entry.

By default, explanations are deterministic templates built only from the row's own fields. Every template names each triggered rule and uses the row's real amount, date, vendor, and people.

Example: "ENT-0126 shows 1 red flag: Self-Approval. Hamza Sheikh both posted and approved the Rs. 4,34,442.46 payment. ..."

11. Groq explanations (optional)

Groq is **off by default**.

1. Open **Groq API keys** in the sidebar.
2. Paste one or more keys (one per line) and press **Add key**.
3. Press **Test** beside a key to check it, **Remove** to delete one, or **Clear all keys**.
4. Choose a model and press **Refresh models** to reload the list.
5. Turn on **Use LLM explanations (Groq)**.

Safety rules:

? Keys live **only** in `st.session_state`. They are never written to disk, logs, Excel, CSV, or error messages. The UI shows masked keys such as `gsk_...abcd`.

? Only structured facts are sent: flags, amounts in rupees, dates in `MM/DD/YYYY`, vendor, poster, approver, risk score, and reasons.

? The system prompt says: *Use only these facts; do not invent amounts, names or dates; 3 sentences max; clear simple English.*

? If the reply contains a number that is not in the row facts, the reply is discarded and the template is used.

? On `429`, `401`, `403`, timeout, or network failure, the client moves to the next key. If every key fails, the app shows a notice and uses templates.

? Requests time out after 20 seconds; `temperature=0.2`, `max_tokens=220`.

? Replies are cached per entry, model, and language.

? Keys may also come from the optional `GROQ_API_KEYS` environment variable or `.env` file; they are loaded into the session and still never written anywhere.

Example: with two keys and the first one rate-limited, the first request

automatically uses the second key and the UI shows the model that produced the text: `Explanation source: Groq (llama-3.3-70b-versatile)`.

12. Calendar tab

The Calendar tab follows the selected financial year (July–June by default).

? **Month heatmap:** each day is coloured by flagged volume; weekends are shaded; holidays show a starred hover tooltip; Critical days get a red outline.
? **Year heatmap:** a GitHub-style grid for the whole year.
? **Insight cards:** weekend/holiday flags, busiest day, the share of flags in the last three working days of each month versus the rest, and the top month by risk.
? **Day inspector:** pick a date to list its entries with risk badges and explanations.

Example: in the demo, Saturday 11/01/2025 contains ENT-0601, a Critical MultiRule entry. Selecting that date shows its badges and explanation.

A zero-flag ledger still renders an empty calendar safely.

13. Export & Accuracy tab

Excel report

Five sheets:

1. **Summary** — KPIs, live formulas, rule-hit bar chart, risk-mix pie chart.
2. **Flagged Entries** — filters, risk fills, `mm/dd/yyyy` dates, `"Rs. "#,##0.00` amounts.
3. **Flag Breakdown** — counts and amounts by rule and vendor.
4. **AI Explanations** — one explanation per flagged entry.
5. **Methodology** — rule logic, weights, scoring, ISA 240, and Karachi context.

The header reads **"AuditGuard Audit Report — Karachi"** and includes **"Prepared in Karachi, Pakistan"**.

CSV

The CSV contains flagged rows with `Amount (Rs.)`, `MM/DD/YYYY` dates, risk level, flags, and reasons. It opens cleanly in Excel because it is UTF-8 with BOM.

Accuracy

With a `GroundTruth` column:

? overall TP, FP, FN, TN, precision, recall, F1, and accuracy,

? a confusion-matrix heatmap,

? per-rule precision/recall/support,

? missed-anomaly and false-alarm tables.

A `MultiRule` row counts as detected only when at least two flags fire.

Per-rule metrics use one-vs-rest over the six single-rule labels.

Example: the demo scores 57 TP, 0 FP, 0 FN, 549 TN → 100% precision, recall, and F1.

14. Privacy and safety

? No database and no server-side persistence.

? Uploaded files and Groq keys disappear when the session ends.

? Excel text that starts with `=`, `+`, `-`, `@`, tab, or newline is forced to plain text to prevent formula injection.

? The LLM never changes a flag, score, or accuracy number.

? An ungrounded LLM reply is rejected automatically.

15. Troubleshooting

16. Demo data at a glance

The audit flags **57** entries: 41 Medium, 4 High, 2 Critical, and 10 Low.

The demo includes 25 Karachi vendors, 10 accounts, and 6 staff members.

10. Demo walkthrough

AuditGuard — FinTech Expo Demo Script

Target length: 5–6 minutes. **Presenter:** one speaker, one laptop.

Audience: judges, finance faculty, auditors, students.

AuditGuard is presented as a Karachi-built audit dashboard: rupees, `MM/DD/YYYY` dates, Pakistan holidays, a July–June financial year, and an optional Groq explanation layer that never sees the full ledger.

Before you present (5-minute checklist)

1. Start the app:

```text
cd "D:\AI Audit Red-Flag Dashboard\auditguard"
..\.venv\Scripts\activate
python -m streamlit run app.py
```

2. Open `http://localhost:8501` in Chrome at 100% zoom.
3. Click **Load Demo Data** once to warm the cache, then reload the page so the demo starts from a clean landing state.
4. Keep `data\AuditGuard_Karachi_Dataset_blind.xlsx` nearby for the upload step.
5. Keep `data\AuditGuard_Karachi_Dataset.xlsx` available for the accuracy step, because it contains `GroundTruth`.
6. Have a screenshot of the **Flagged Entries** tab, the **Calendar** tab, and a downloaded Excel report ready in case the projector or Wi-Fi fails. The app runs locally.
7. Close other browser tabs and notifications.
8. Leave the Groq toggle **off** unless the key is tested and the venue Wi-Fi is reliable.

0:00–0:45 | The problem

"Every month, finance teams receive thousands of journal entries. Auditors do not struggle to find data; they struggle to find the handful of entries worth investigating. Sampling catches some errors, but it is slow, inconsistent, and hard to explain to a client.

>

AuditGuard turns a general ledger into a ranked, explainable review list in seconds. It applies six audit tests, shows the evidence behind every flag, plots risky days on a calendar, exports a workpaper-ready Excel pack, and proves its own precision and recall when labelled data is available.

>

It is built for Pakistan: rupees, `MM/DD/YYYY` dates, Karachi time, and the Pakistani weekend and public holidays."

Show the landing page: schema chips, rupee labels, and empty states.

0:45–1:30 | Upload and run

? Point to the sidebar schema list: `EntryID`, `Date`, `Account`, `Vendor`, `Description`, `Amount`, `PostedBy`, `ApprovedBy`.

? Upload `AuditGuard_Karachi_Dataset_blind.xlsx` (no `GroundTruth`).

? Say: "The file is validated in the session. Amounts are in rupees, dates are

read as `MM/DD/YYYY`, and bad rows are reported rather than silently dropped. There is no database and no external upload."

? Press **Run Audit**.

? Expected result: toast reports **57 flagged of 606 entries**; the hero status becomes **Audit complete**.

If upload misbehaves, click **Load Demo Data** and continue.

1:30–2:15 | Overview

? Four KPI cards: total entries, flagged entries and percentage, amount at risk in rupees, overall risk score/level.

? Rule-hit bar chart: duplicate, round, weekend, just-below-limit, self-approval, Benford.

? Risk donut and flagged timeline: red markers cluster on the anomaly dates.

? Benford chart: observed first digits versus the expected distribution, with a p-value callout.

? Say: "Every score is arithmetic, not a black box. The weights are visible and editable under Advanced settings, and the amounts are in lakh/crore format so they read naturally in Pakistan."

Expected result: **57 flagged**, **Rs. 77,29,596.13 at risk**, overall risk **47.3 / 100 (High)**. Benford produces the most hits (18), followed by duplicates (12) and self-approvals (11).

2:15–3:05 | Investigate a flagged entry

? Open **Flagged Entries**.

? Point out the filters: risk level, flag type, vendor, minimum score, date range, description search.

? Select **Self-Approval** in **Flag type**; the table narrows to 11 entries.

? Pick a top entry. Read the detail card:

? every field with `MM/DD/YYYY` dates and `Rs.` amounts,

? the numeric reason ("Posted and approved by the same person…"),

? the plain-English explanation,

? recommended audit procedures.

? Switch **Flag type** to **Duplicate**, open one entry, and show the **Related duplicate entry** card with the earlier `EntryID`. If the shared description includes `INV-2025-0142`, point out that the invoice reference is quoted as extra evidence.

? Say: "The auditor gets the control that failed, the amount, the people, and the next procedure — not just a red dot."

Expected result: **11 of 57** for Self-Approval; duplicate detail names both postings, the day gap, and any shared invoice reference.

3:05–3:35 | Calendar

? Open **Calendar**.

? The financial year defaults to **2025–26 (Jul–Jun)** and the month selector opens on December 2025.

? Show the month heatmap: weekends shaded, holiday tooltips on hover, Critical days with a red outline.

? Point at 11/01/2025: ENT-0601 is a Critical MultiRule entry.

? Scroll to the year heatmap and the four insight cards.

? Pick a date in **Inspect a day** and show the entry badges and explanations.

? Say: "Auditors think in months and cut-offs. The calendar turns the same flags into a visual review plan and highlights clustering near month-end and weekends."

3:35–4:10 | AI Insights and Groq

? Open **AI Insights**.

? Read the executive summary: 606 entries, 57 flagged, rupee exposure, dominant pattern, riskiest vendor, and the top entry to start with.

? Show the top-N cards and the **Explain any flagged entry** selector.

? Say: "The default explanation is a deterministic template built only from the row's own fields. The optional Groq switch only rewrites the wording; it never changes a flag or sends the ledger. Keys stay in this session only, are rotated automatically if one fails, and any invented number is rejected. On any error we fall back to the template."

Do **not** enable Groq during the live demo unless the key is already tested and the venue Wi-Fi is reliable.

4:10–4:50 | Export and accuracy

? Open **Export & Accuracy**.

? Download the Excel report and open it. Show:

? **Summary** with live formulas and native charts,

? **Flagged Entries** with filters, risk fills, `mm/dd/yyyy` dates, and `Rs.` amounts,

? **AI Explanations** and **Methodology** (including ISA 240 and the Karachi report header).

? Say: "This is not a raw CSV dump; it is a workpaper-ready pack with the rule logic documented inside the file."

? Scroll to **Detection accuracy**. Show precision, recall, F1, the confusion matrix, and the per-type table.

? Say: "On the labelled demo: 57 of 57 anomalies caught, no false alarms, 100% precision and 100% recall. MultiRule rows count as detected only when at least two flags fire, so the perfect score is earned, not assumed."

Expected result: TP 57, FP 0, FN 0, precision 100%, recall 100%, F1 100%.

4:50–5:25 | Market and business model

Use illustrative numbers and label them as assumptions.

"Mid-market finance teams and audit firms still rely on spreadsheet sampling. A typical audit-analytics seat is priced in the low millions of rupees per year; our wedge is a lightweight, explainable review layer that sits on top of the exports teams already have.

>

Illustrative model: **Rs. 2.4 million per year per audit team**, 50 teams in year two gives **Rs. 120 million ARR**. Gross margin is high because processing is local and the core engine is deterministic. Land with a free single-ledger audit, expand to multi-entity and reviewer workflow, then integrate with ERP exports.

>

The moat is not a proprietary model; it is the combination of audit-specific rules, Karachi localization, explainability, accuracy reporting, and a workpaper-ready export that a finance team can adopt without a data-science hire."

Keep this section conversational; judges care more about clarity than precise market-size figures.

5:25–5:40 | Close

"AuditGuard combines accounting controls, statistics, explainable rules, Excel reporting, and a usable dashboard, localized for Pakistan. It is decision support, not an audit opinion: the auditor stays in control, every flag is explainable, and the app reports its own mistakes. Thank you — questions?"

Fallback plan

Judge Q&A (12 likely questions)

1. Is this machine learning?

No. The detection is deterministic rules plus a chi-square Benford test. That is a deliberate choice: auditors need to explain and defend every finding. The optional Groq layer only rewrites explanations and never changes a flag.

2. What about false positives?

The accuracy tab reports them directly. On the demo, the labels are complete, so there are zero false alarms; the vendor-level false-alarm card remains visible for future labelled data where a whole vendor is not manipulated. Review and risk ranking are designed for triage, not automatic accusation.

3. How is this different from Excel?

Excel can hold the data, but it cannot apply six consistent tests, score and rank entries, generate grounded explanations, enforce formula-injection safety, and report precision/recall. AuditGuard also exports back to Excel, so it augments the auditor's existing tool rather than replacing it.

4. How do you make money?

Illustratively: a per-team annual subscription (around Rs. 2.4 million), a free single-ledger tier for land-and-expand, and later add-ons for multi-entity consolidation, reviewer workflow, and ERP connectors. The demo does not claim validated pricing.

5. Can the LLM hallucinate amounts or names?

Groq receives only structured facts and is instructed to use only those facts in at most three sentences. If the reply contains a number that is not in the row facts, it is discarded and the template is used. Flags, scores, and accuracy never depend on the LLM.

6. Where does my data go?

Nowhere by default. Processing is local to the Streamlit session; there is no database and no persistence. Uploads disappear when the session ends. Only if a user explicitly enables Groq and supplies a key are structured facts sent to Groq; keys live only in the session and are never stored or logged.

7. Why Benford's Law?

It is a population-level screen for fabricated or manipulated amounts. We apply it per vendor with a minimum of 15 entries and $p < 0.01$, then flag the whole vendor. It is a pointer for sampling, not proof of fraud. Rent from Pearl Property Rentals is excluded from digit testing.

8. Why these weights?

They reflect control severity: self-approval breaks segregation of duties (35),

followed by duplicates (30), threshold structuring (25), Benford (20), round amounts (15), and weekend postings (10). Every weight is editable, and the score is a capped sum, so the math is auditable. The six MultiRule demo rows combine two or three rules and land in High or Critical.

9. What happens with 20,000 rows?
The audit completes in roughly three seconds on a normal laptop. The app was tested at that size; the table is capped and filtered client-side. Larger deployments would move scoring to a background job and a database.

10. How do you handle missing or messy columns?
The app blocks only when the ledger cannot be used: missing required columns, all dates unreadable, all amounts non-numeric, negative amounts, or fewer than two rows. Partial issues become warnings, and the affected rules skip only the rows they cannot evaluate. A column-mapping panel handles common header aliases.

11. What is the biggest limitation?
Duplicate matching uses vendor + amount + a seven-day window because the fixed schema has no invoice field. Invoice references in descriptions are supporting evidence, not a key. Benford also produces vendor-level false alarms by design.

12. What is next?
Invoice and PO matching as first-class keys, configurable rule sets per engagement, reviewer sign-off and an immutable audit trail, multi-sheet or multi-entity consolidation, PDF export, and ERP connectors. Production also needs SSO, role-based access, and monitoring.

One-sentence takeaway

"AuditGuard gives auditors a fast, explainable, and measurable first pass over a general ledger — in rupees, on Karachi time, and with its work shown."

11. Multi-year holidays

Fixed Pakistan holidays are generated for every year detected in the uploaded Date column. If a file covers 2024 through 2027, fixed holidays are added for 2024, 2025, 2026, and 2027. Extra/moon-sighting dates are exact manually entered dates; the demo prefill contains 2025 suggestions only and does not calculate future moon-sighting dates. Verify Islamic dates before real use.

12. Common commands quick reference

```
Start existing environment: cd "D:\AI Audit Red-Flag Dashboard\auditguard" ..\.venv\Scripts\activate
python -m streamlit run app.py Install/restore packages: pip install -r requirements.txt Run tests:
python -m pytest -q Stop: Ctrl+C in the Streamlit terminal
```

13. Troubleshooting

If localhost:8501 does not open, check the terminal for an error and open the URL manually. If the port is busy, stop the previous Streamlit process or use port 8502. If dates look swapped, select the correct input date format. If columns are missing, use mapping. If Groq fails, use deterministic templates. If filters show zero rows, reset risk levels, flag type, vendor, search, date range, and minimum score.

14. Industry usefulness and limitations

AuditGuard helps audit firms, finance teams, internal audit departments, and SMEs prioritize review over existing spreadsheet exports. It is not proof of fraud. The documented target is about 20,000 rows in roughly three seconds on the development laptop; very large files increase memory, duplicate grouping, rendering, Benford, and Excel export time. Production deployment would add authentication, role-based access, background jobs, database persistence, reviewer workflow, and monitoring.